



BCA Ist Semester

Fundamentals of Operating system

Course Code: BCAC0006

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Shell Layer Architecture

- The functioning of OS is done in three levels.
- On the outer crust reside the application programs and other utilities.
- At the heart of OS is the Kernel, which interacts with the actual hardware in machine language.
- Between these two lies the **Shell** which is responsible for the communication b/w the two layers.

Tools and Applications

Shell

Kernel

Hardware

Structure of Operating system

- The Operating System is divided into three main entities:

- **1) The Kernel** It's the core of the system. It controls all the tasks, schedules all the processes and carries out all the crucial features of the operating system.
- It manages the files, carries out all the data transfer between the file system and the hardware, and also manages memory.
- Among major duties of the Kernel is to keep track of the programs that are executing, allot processor time to each and also decide when one program stops and another starts.
- The Kernel also handles information exchange between the computer and its terminals, tapes drives and printers.

2) The Shell

- It is the Command Interpreter of the operating system. It accepts commands from the user and analyses and interprets these commands and finally starts executing their appropriate executable file.
- It then requests the Kernel to carry out the actual transfer of data which finally leads to the output that is displayed on the terminal.
- Hence the Shell act as an Middleman between the Kernel and the User of the Operating System.
- There are three shells mainly used in UNIX.

- **i) The Bourne Shell**
- This is one of the most widely used shells in the Unix world. It was developed by Steve Bourne of AT&T Bell Labs in the late 1970's. It is the primary Unix Command Interpreter and comes along with every Unix System. The \$ prompt is the trademark of Bourne Shell.
- **ii) The C Shell**
- It was created by Bill Joy at University of California, Berkeley. It has 2 advantages over the Bourne Shell:

- Aliasing. The C shell permits you to call frequently used commands by your own formulated abbreviations. This proves very useful when lengthy commands which are used time and again are renamed by you. Instead of typing the entire command you can simply use the short alias at the command line.
- A history mechanism. Previously typed command can be recalled, since the C Shell keeps track of all the command issued at the command line. This feature is similar to the one provided by the program DOSKEY in MS-DOS.

- **iii) The Korn Shell**
- Developed by David Korn of AT&T, this shell was designed to be much bigger than the Bourne Shell and includes several features that make it more superior.
- It is more powerful, versatile and efficient than Bourne Shell.
- The Korn Shell includes all the enhancements in the C shell, like command history and aliasing, and offers a more few features itself.

- **3) Tools and Application** The outermost layer of the operating system is its tools and applications.
- As some versions of Unix are decked with more than 400 tools and applications.
- These tools can be invoked from the command line itself and help to perform day-to-day complex tasks of the system.
- These are placed one level above the Shell and can be extended as per user convenience.